

STAT 212: Final Project

Noise Stability on the Boolean Hypercube via
a Renormalized Brownian Motion

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1 Introduction

1.1 Motivation

The paper studies Boolean functions $f : C_n \rightarrow \{0, 1\}$, where $C_n = \{-1, 1\}^n$ and μ_n is the uniform measure on C_n . A central question is how much information about $f(X)$ survives after the input is corrupted by noise. For $\rho \in [0, 1]$ and $x \in C_n$, let $N_{\rho, x}$ be the random vector in C_n with independent coordinates satisfying $\mathbb{E}[(N_{\rho, x})_i] = \rho x_i$.

Equivalently, each coordinate stays the same with probability $(1 + \rho)/2$ and flips with probability $(1 - \rho)/2$. The noise operator is $T_\rho[f](x) = \mathbb{E}[f(N_{\rho, x})]$. We also define noise stability,

$$\text{Stab}_\rho(f) = \int_{C_n} f T_\rho[f] d\mu_n.$$

This measures how well the value of f survives after the input is randomly perturbed.

One of the central conjectures in this area is the Courtade–Kumar conjecture. If X and Y are random variables, their mutual information is $I(X; Y) = H(X) - H(X | Y)$, where H is Shannon entropy. The conjecture is as follows.

Conjecture 1 (Courtade–Kumar). *Let X be uniformly distributed on C_n . Among all Boolean functions $f : C_n \rightarrow \{0, 1\}$, the function which maximizes $I(X; f(N_{\rho, X}))$ is the dictator function.*

A dictator function depends on only one coordinate, for example $f(x) = 1_{\{x_1 > 0\}}$. In Sections 1–3, the paper introduces a new noise model, defined through a stochastic process, for which an analogue of the conjecture can be proved.

2 Key Findings

2.1 Preliminaries

The first key step is to pass from the discrete cube C_n to $[-1, 1]^n$ using the multilinear extension of f . If

$$f(x) = \sum_{A \subseteq [n]} \widehat{f}(A) \chi_A(x), \quad \chi_A(x) = \prod_{i \in A} x_i,$$

then the same polynomial can be evaluated on $[-1, 1]^n$. In what follows we use the same notation f for this multilinear extension. With this extension,

$$T_\rho[f](x) = \sum_{A \subseteq [n]} \widehat{f}(A) \rho^{|A|} \chi_A(x) = \sum_{A \subseteq [n]} \widehat{f}(A) \chi_A(\rho x) = f(\rho x).$$

So once f is viewed on the continuous cube, adding noise is the same as scaling the input by ρ . This leads to the generalized stability functional

$$S_{\varphi, \nu}(f) = \int_{[-1, 1]^n} \varphi(f(x)) d\nu^n(x) - \varphi\left(\int_{[-1, 1]^n} f d\nu^n\right),$$

where $\varphi : [0, 1] \rightarrow \mathbb{R}$ is convex and ν is a symmetric probability measure on $[-1, 1]$.

2.2 Stochastic constructions

We now summarize the renormalized Brownian motion construction and explain why it is the right process for this problem. Instead of introducing noise by flipping bits all at once, the paper lets the bits emerge gradually through a Brownian process. In one dimension, we start at 0 and consider

$$X(0) = 0, \quad dX(t) = \sqrt{(1 + X(t))(1 - X(t))} \mathbf{1}_{|X(t)| \leq 1} dB(t).$$

Since $(1 + X(t))(1 - X(t)) = 1 - X(t)^2$, the diffusion slows down as the process approaches ± 1 , and once it reaches the boundary it stops. So this gives a continuous way of sampling a random bit.

In higher dimensions, we run the same construction independently in each coordinate: $X(0) = 0$ and $dX(t) = \sigma_t dB(t)$, where σ_t is diagonal and $(\sigma_t)_{i,i} = \sqrt{(1 + X_i(t))(1 - X_i(t))} \mathbf{1}_{|X_i(t)| \leq 1}$. We write $\nu_t = \text{law}(X_1(t))$ for the law of one coordinate at time t .

Now fix $f : C_n \rightarrow [0, 1]$ and define $N_t = f(X(t))$. This is the main process in the argument. Since f is multilinear, it has the form $f(x) = \sum_{A \subseteq [n]} \hat{f}(A) \prod_{i \in A} x_i$, so it is enough to understand one monomial $\prod_{i \in A} X_i(t)$. Each coordinate $X_i(t)$ is itself a martingale, and different coordinates have zero cross-variation because they are driven by independent Brownian motions. Applying Itô's formula to a product therefore produces no drift term, so each monomial is a martingale. Since N_t is a linear combination of these monomials, it follows that $N_t = f(X(t))$ is also a martingale.

Now, the quantities of interest are expressed directly in terms of N_t . Namely,

$$S_{\varphi, \nu_t}(f) = \mathbb{E}[\varphi(N_t)] - \varphi\left(\int_{C_n} f d\mu_n\right),$$

so, once the mean $\int_{C_n} f d\mu_n$ is fixed, maximizing $S_{\varphi, \nu_t}(f)$ is equivalent to maximizing $\mathbb{E}[\varphi(N_t)]$.

2.3 A variant of Courtade–Kumar

The main result is the following.

Theorem 1. *Let $\varphi : [0, 1] \rightarrow \mathbb{R}$ be convex and $t \in (0, \infty)$. Among all Boolean functions $f : C_n \rightarrow \{0, 1\}$, the function which maximizes the quantity S_{φ, ν_t} is the dictator function.*

This does not prove Conjecture 1. The reason is that Theorem 1 is proved for the functional S_{φ, ν_t} , where the averaging measure is the Brownian measure ν_t , while Conjecture 1 is about mutual information in the original classical noise model. So even though both statements identify the dictator as extremal, the quantity being maximized is not the same.

To sketch the main idea, we restrict to the balanced case. The general case is handled in the same way after fixing the mean $\int f d\mu_n$, since the second term in $S_{\varphi, \nu_t}(f)$ then becomes constant.

Sketch of proof for Theorem 1. Let $f(x) = \mathbb{1}_{\{x_i > 0\}}$ be a dictator, and let $g : C_n \rightarrow \{0, 1\}$ be a balanced Boolean function, so $\int g d\mu_n = 1/2$. Define $M_t = f(X(t))$ and $N_t = g(\tilde{X}(t))$, where $\tilde{X}(t) \stackrel{d}{=} X(t)$ but is defined on a different probability space.

A key proposition says that if $\{M_t : t \geq 0\}$ and $\{N_t : t \geq 0\}$ are continuous martingales with $M_0 = N_0$, then they can be realized on the same probability space together with a standard Brownian motion $\{W_\tau : \tau \geq 0\}$ such that

$$M_t = W_{[M]_t}, \quad N_t = W_{[N]_t}, \quad \forall t \geq 0.$$

So both martingales can be viewed as time changes of the same Brownian motion, and the problem reduces to comparing their quadratic variations.

Let $\mathcal{T}_1, \mathcal{T}_2$ be the inverse functions of the non-decreasing maps $t \mapsto [M]_t$ and $t \mapsto [N]_t$. Then $M_{\mathcal{T}_1(\tau)} = N_{\mathcal{T}_2(\tau)} = W_\tau$ for $0 \leq \tau \leq \tau_{\max}$, where $\tau_{\max} = \min\{t \geq 0 : W_t \in \{0, 1\}\}$ is a stopping time.

The main claim is that

$$\begin{aligned} \frac{d}{dt}[M]_t &= \|\sigma_t \nabla f(X(t))\|_2^2 = f(X(t))(1 - f(X(t))) = M_t(1 - M_t), \\ \frac{d}{dt}[N]_t &= \|\sigma_t \nabla g(\tilde{X}(t))\|_2^2 \leq g(\tilde{X}(t))(1 - g(\tilde{X}(t))) = N_t(1 - N_t). \end{aligned} \quad (1)$$

For the dictator, the first identity comes from writing $f(x) = (x_i + 1)/2$. The second follows from Itô's formula together with Parseval's inequality in the paper.

Now along the common Brownian path we have $M_{\mathcal{T}_1(\tau)} = N_{\mathcal{T}_2(\tau)} = W_\tau$. Substituting this into (1) gives

$$\mathcal{T}'_1(\tau) = \frac{1}{W_\tau(1 - W_\tau)} \leq \mathcal{T}'_2(\tau)$$

for $0 \leq \tau \leq \tau_{\max}$. Integrating, we obtain $\mathcal{T}_1(\tau) \leq \mathcal{T}_2(\tau)$.

Now fix $t_0 \geq 0$, and let $T_1 = [M]_{t_0}$ and $T_2 = [N]_{t_0}$. Then $T_1 \geq T_2$. Since W_t is a martingale, conditioning on W_{T_2} , $\mathbb{E}[W_{T_1} \mid W_{T_2}] = W_{T_2}$. Applying Jensen's inequality to the convex function φ , we obtain

$$\mathbb{E}[\varphi(f(X(t_0)))] = \mathbb{E}[\varphi(W_{T_1})] = \mathbb{E}[\mathbb{E}[\varphi(W_{T_1}) \mid W_{T_2}]] \geq \mathbb{E}[\varphi(W_{T_2})] = \mathbb{E}[\varphi(g(\tilde{X}(t_0)))].$$

Thus, among balanced Boolean functions, the dictator maximizes S_{φ, ν_t} , and as noted before applying the same argument applies more generally after fixing the mean $\int f d\mu_n$ yields the full theorem. $\square$

3 Conclusions

The main conclusion is that dictators maximize S_{φ, ν_t} for every convex φ , giving a natural variant of the Courtade–Kumar conjecture. After this, the paper turns to the Majority is Stablest theorem. Using the same stochastic framework, it gives a new proof of the theorem and also improves the quantitative error term, with polynomial rather than logarithmic dependence on the maximal influence.